A/B Test Analysis & Experimentation Dashboard

Project Overview

This project was developed to analyze and evaluate the effectiveness of two landing page variations using statistical experimentation techniques. The project combines SQL analysis, statistical interpretation, and Power BI visualization to measure user conversion behavior and provide business-oriented insights.

Objectives

• Compare conversion rates between control and treatment groups.
• Measure relative conversion lift.
• Analyze weekly user behavior trends.
• Evaluate statistical significance.
• Estimate projected business impact.
• Build an interactive Power BI dashboard.

Tools and Technologies Used

• SQL (MySQL)
• Power BI
• DAX
• Statistical Analysis
• Excel / CSV Dataset

Dataset Description

The dataset contains user-level A/B testing data collected during an online experiment.

Columns used:
• user_id
• timestamp
• group
• landing_page
• converted
• week

Methodology

1. Data Cleaning and Validation
2. SQL-based Exploratory Analysis

3. Conversion Rate Computation
4. Relative Lift Calculation
5. Statistical Testing Interpretation
6. Revenue Impact Estimation
7. Dashboard Development in Power BI
8. Final Experiment Interpretation

Key Statistical Metrics

• Control Conversion Rate: 12.04%
• Treatment Conversion Rate: 11.88%
• Relative Conversion Lift: -1.31%
• P-Value: 0.19
• Z-Statistic: -1.31
• Statistical Power: 0.80
• Experiment Verdict: Do Not Ship

Statistical Interpretation

The treatment landing page showed slightly lower conversion performance compared to the control landing page. The p-value of 0.19 is greater than the significance threshold of 0.05, indicating that the observed difference was not statistically significant. Based on the observed performance decline and lack of statistical significance, deployment of the treatment landing page is not recommended.

Power BI Dashboard

Page 1 focuses on KPI overview, conversion comparison, user distribution, projected revenue, and experiment verdict.

Page 2 focuses on statistical validation, weekly trend analysis, user allocation consistency, and final experiment interpretation.

Key DAX Measures

• Total Users
• Total Conversions
• Control Conversion Rate
• Treatment Conversion Rate
• Relative Conversion Lift

Assumptions and Analytical Considerations

• Users were randomly and independently assigned to experiment groups.
• A fixed projected revenue value of $500 per conversion was assumed.

• Revenue metrics are projected estimates and not actual business revenue.
• External marketing or seasonal effects were assumed minimal.
• Statistical significance threshold (α) of 0.05 was used.
• Weekly aggregation was considered sufficient for trend analysis.
• The dataset was assumed to represent normal platform behavior.

Key Findings

• The treatment landing page underperformed compared to the control page.
• The observed conversion decline was approximately 1.31%.
• Statistical evidence was insufficient to support deployment.
• Weekly trend analysis showed no sustained treatment improvement.
• User distribution remained balanced across experiment groups.

Conclusion

This project successfully demonstrated the complete workflow of an A/B testing and experimentation analysis pipeline using SQL, Power BI, DAX, and statistical interpretation techniques.

The analysis showed that the treatment landing page failed to outperform the control landing page and did not achieve statistical significance. Therefore, deployment of the treatment variation was not recommended.

Future Improvements

• Real-time experimentation dashboard integration
• Confidence interval visualization
• Bayesian experimentation analysis
• Real transaction data integration
• Multi-variant experimentation support